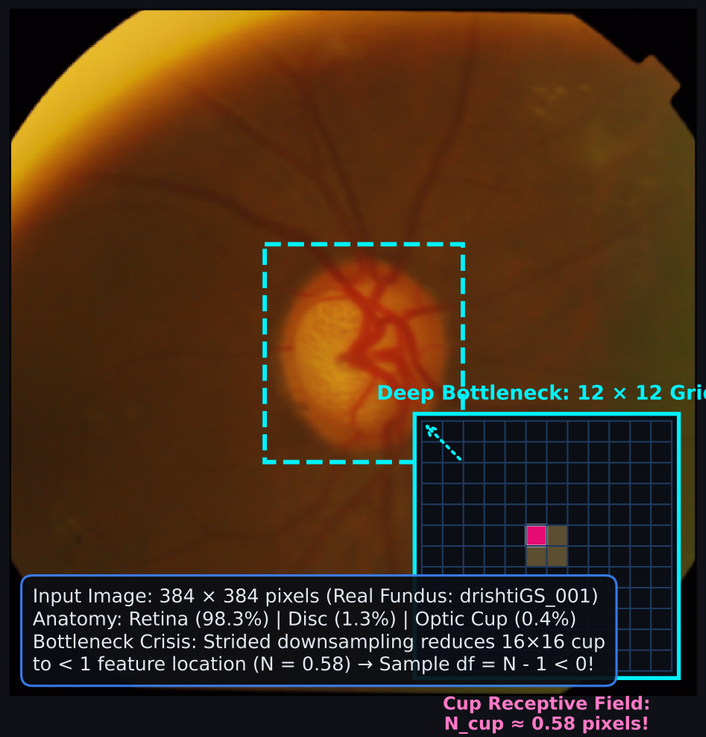
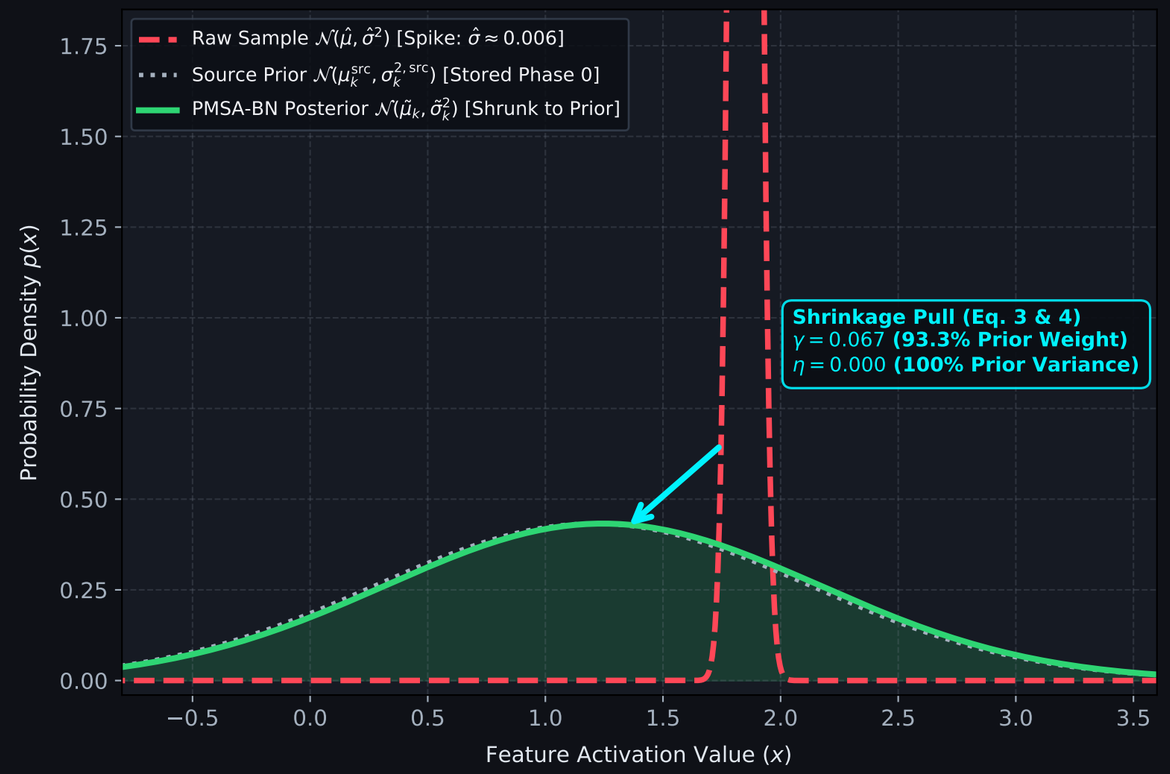
Empirical Bayes Shrinkage (Eq. 3 & 4): Visual Proof of Numerical Stability at $B = 1$
Evaluated on Real Retinal Fundus (Drishti-GS Test Image drishtiGS_001)

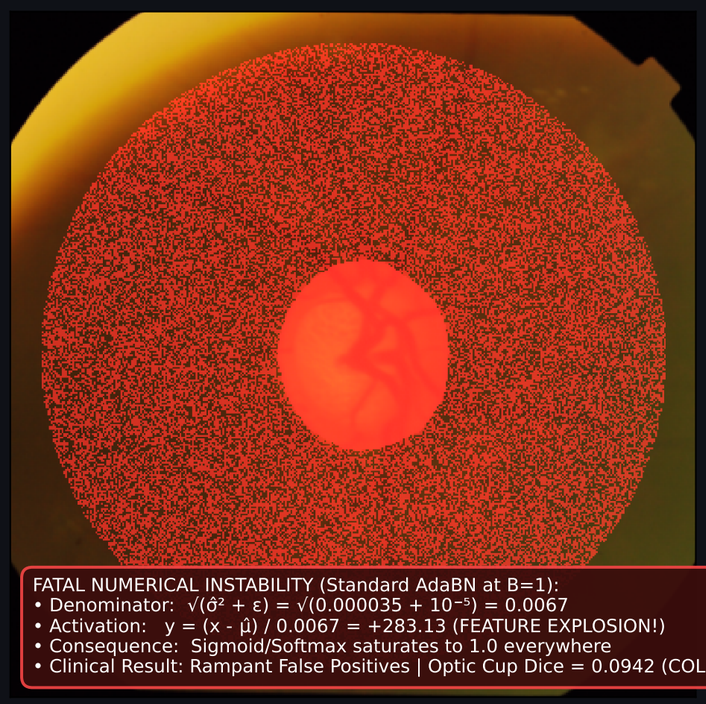
(A) The Spatial Dilemma: High-Res Input vs. Deep Bottleneck



(B) Bayesian Shrinkage Density in Activation Space



(C) Test-Time Adaptation WITHOUT Shrinkage ($\kappa = 0$, Raw AdaBN at $B = 1$)



(D) Test-Time Adaptation WITH PMSA-BN Shrinkage ($\kappa_\mu = 8, \kappa_\sigma = 16$)

